Sorting & Searching

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What is sorting?

- Sorting refers to shuffling a sequence of objects by a comparison operator <.
- The operator tells whether a < b is true.
- The operator must satisfy the following for any elements a, b, c in context.
  - Irreflexivity: a < a is false.
  - Asymmetric: If a < b is true, then b < a is false.
  - Transitive: If a < b and b < c is true, then a < c is true.
  - Connectedness: If a != b, then either a < b or b < a is true.

Sorting Algorithms

- Bubble Sort
- Selection Sort
- Counting Sort
- Merge Sort
- Radix Sort

https://codeforces.com/problemset/problem/1851/B

You have an array of integers a of length n. You can apply the following operation to the given array:

- Swap two elements a_i and a_j such that $i \neq j$, a_i and a_j are either **both** even or **both** odd.

Determine whether it is possible to sort the array in non-decreasing order by performing the operation any number of times (possibly zero).

For example, let $a = [7, 10, 1, 3, 2]$. Then we can perform 3 operations to sort the array:

1. Swap $a_3 = 1$ and $a_1 = 7$, since 1 and 7 are odd. We get $a = [1, 10, 7, 3, 2]$;
2. Swap $a_2 = 10$ and $a_5 = 2$, since 10 and 2 are even. We get $a = [1, 2, 7, 3, 10]$;
3. Swap $a_4 = 3$ and $a_3 = 7$, since 3 and 7 are odd. We get $a = [1, 2, 3, 7, 10]$.

Input

The first line of input data contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The description of the test cases follows.

The first line of each test case contains one integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of array a.

The second line of each test case contains exactly n positive integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of array a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

https://codeforces.com/problemset/problem/1312/B

You are given an array $a_1, a_2, \ldots, a_n$. Array is good if for each pair of indexes $i < j$ the condition $j - a_j \neq i - a_i$ holds. Can you shuffle this array so that it becomes good? To shuffle an array means to reorder its elements arbitrarily (leaving the initial order is also an option).

For example, if $a = [1, 1, 3, 5]$, then shuffled arrays $[1, 3, 5, 1]$, $[3, 5, 1, 1]$ and $[5, 3, 1, 1]$ are good, but shuffled arrays $[3, 1, 5, 1]$, $[1, 1, 3, 5]$ and $[1, 1, 5, 3]$ aren't.

It's guaranteed that it's always possible to shuffle an array to meet this condition.

Input

The first line contains one integer t ($1 \leq t \leq 100$) — the number of test cases.

The first line of each test case contains one integer n ($1 \leq n \leq 100$) — the length of array a.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 100$).

Output

For each test case print the shuffled version of the array a which is good.

Example

input Copy

```
3
1
7
4
1 1 3 5
6
3 2 1 5 6 4
```

output Copy

```
7
1 5 1 3
2 4 6 1 3 5
```

Custom Sorting

```
19  struct box {
20      int a, b;
21      bool operator<(box other) const {
22          if (a != other.a)
23              return a < other.a;
24          return b < other.b;
25      }
26  };
27
28  int main() {
29      ios_base::sync_with_stdio(0); cin.tie(0); cout.tie(0);
30
31      vector<box> v;
32
33      // populate contents of v
34
35      sort(v.begin(), v.end());
36      return 0;
37  }
```

Custom Sorting

```
19  struct box {
20      int a, b;
21  };
22
23  int main() {
24      ios_base::sync_with_stdio(0); cin.tie(0); cout.tie(0);
25
26      vector<box> v;
27      // populate contents of v
28
29      vector<int> ord(sz(v));
30      iota(v.begin(), v.end(), 0);
31
32      sort(ord.begin(), ord.end(), [&](int i, int j) {
33          return make_pair(v[i].a, v[i].b) < make_pair(v[j].a, v[j].b);
34      });
35
36      for (int i : ord) {
37          // now v[i] is the ith element in sorted order
38      }
```

https://cses.fi/problemset/task/1084

There are n applicants and m free apartments. Your task is to distribute the apartments so that as many applicants as possible will get an apartment.

Each applicant has a desired apartment size, and they will accept any apartment whose size is close enough to the desired size.

Input

The first input line has three integers n, m, and k: the number of applicants, the number of apartments, and the maximum allowed difference.

The next line contains n integers $a_1, a_2, \ldots, a_n$: the desired apartment size of each applicant. If the desired size of an applicant is x, they will accept any apartment whose size is between $x - k$ and $x + k$.

The last line contains m integers $b_1, b_2, \ldots, b_m$: the size of each apartment.

Output

Print one integer: the number of applicants who will get an apartment.

Constraints

- $1 \le n, m \le 2 \cdot 10^5$
- $0 \le k \le 10^9$
- $1 \le a_i, b_i \le 10^9$

Example

Input:

```
4 3 5
60 45 80 60
30 60 75
```

Output:

```
2
```

https://cses.fi/problemset/task/1090

There are n children who want to go to a Ferris wheel, and your task is to find a gondola for each child.

Each gondola may have one or two children in it, and in addition, the total weight in a gondola may not exceed x. You know the weight of every child.

What is the minimum number of gondolas needed for the children?

Input

The first input line contains two integers n and x: the number of children and the maximum allowed weight.

The next line contains n integers $p_1, p_2, \ldots, p_n$: the weight of each child.

Output

Print one integer: the minimum number of gondolas.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x \le 10^9$
- $1 \le p_i \le x$

Example

Input:
```
4 10
7 2 3 9
```

Output:
```
3
```

https://cses.fi/problemset/task/1619

You are given the arrival and leaving times of n customers in a restaurant.

What was the maximum number of customers in the restaurant at any time?

Input

The first input line has an integer n: the number of customers.

After this, there are n lines that describe the customers. Each line has two integers a and b: the arrival and leaving times of a customer.

You may assume that all arrival and leaving times are distinct.

Output

Print one integer: the maximum number of customers.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le a < b \le 10^9$

Example

Input:

```
3
5 8
2 4
3 9
```

Output:

```
2
```

https://cses.fi/problemset/task/1629

In a movie festival n movies will be shown. You know the starting and ending time of each movie. What is the maximum number of movies you can watch entirely?

Input

The first input line has an integer n: the number of movies.

After this, there are n lines that describe the movies. Each line has two integers a and b: the starting and ending times of a movie.

Output

Print one integer: the maximum number of movies.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le a < b \le 10^9$

Example

Input:

```
3
3 5
4 9
5 8
```

Output:

```
2
```

https://cses.fi/problemset/task/1640

You are given an array of n integers, and your task is to find two values (at distinct positions) whose sum is x.

Input

The first input line has two integers n and x: the array size and the target sum.

The second line has n integers $a_1, a_2, \ldots, a_n$: the array values.

Output

Print two integers: the positions of the values. If there are several solutions, you may print any of them. If there are no solutions, print `IMPOSSIBLE`.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x, a_i \le 10^9$

Example

Input:

```
4 8
2 7 5 1
```

Output:

```
2 4
```

https://cses.fi/problemset/task/1074

There are n sticks with some lengths. Your task is to modify the sticks so that each stick has the same length.

You can either lengthen and shorten each stick. Both operations cost x where x is the difference between the new and original length.

What is the minimum total cost?

Input

The first input line contains an integer n: the number of sticks.

Then there are n integers: $p_1, p_2, \ldots, p_n$: the lengths of the sticks.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le p_i \le 10^9$

Example

Input:
```
5
2 3 1 5 2
```

Output:
```
5
```

https://codeforces.com/contest/2122/problem/C

You are given n points (x_i, y_i) on a 2D plane, where n is even. Select $\frac{n}{2}$ disjoint pairs (a_i, b_i) to maximize the sum of Manhattan distances between points in pairs. In other words, maximize

$$\sum_{i=1}^{n/2} |x_{a_i} - x_{b_i}| + |y_{a_i} - y_{b_i}|.$$

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single even integer n ($2 \le n \le 2 \cdot 10^5$) — the number of points.

The i-th of the next n lines contains two integers x_i and y_i ($-10^6 \le x_i, y_i \le 10^6$) — the coordinates of the i-th point.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output $\frac{n}{2}$ lines, the i-th line containing two integers a_i and b_i — the indices of points in the i-th pair.

If there are multiple solutions, print any of them.

Binary Search

- So far we only knew one way to look up an item - Linear Search.
- When the given data is sorted, it lets us search items faster.

Membership problem

- Given n integers, Q queries are asked. In each query you are given x, print whether x is in the array or not.

Lowerbound problem

- Given n integers, Q queries are asked. In each query you are given x.
- Print the smallest number a[i] >= x.

A fit for all situation binary search template

```
int lo = 0, hi = n - 1, ans = -1;
while (lo <= hi) {
    int mid = (lo + hi) / 2;

    if (a[mid] >= x) {
        // write your decisions here
    } else {
        // write your decisions here
    }
}
```

Upperbound problem

- Given n integers, Q queries are asked. In each query you are given x.
- Print the smallest number a[i] > x.

Builtin Functions

```
vector<int> v;
// suppose v is a sorted array

auto iter = lower_bound(v.begin(), v.end(), x);

int l_idx = lower_bound(v.begin(), v.end(), x) - v.begin();
```

https://codeforces.com/edu/course/2/lesson/6/1/practice/contest/283911/problem/D

You are given an array a of n integers $a_1, a_2, \ldots, a_n$.

Your task is to response to the queries like "`How many numbers' values are between` l `and` r?".

Input

The first line of the input contains n — the length of the array ($1 \le n \le 10^5$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($-10^9 \le a_i \le 10^9$).

The third line contains integer k — the number of queries ($1 \le k \le 10^5$).

The following k lines contain a pair of integers l r — query, described above ($-10^9 \le l \le r \le 10^9$).

Output

The output must consist of k integers — responses for the queries.

https://atcoder.jp/contests/abc248/tasks/abc248_d

You are given a sequence of length N: $A = (A_1, \ldots, A_N)$.

Answer Q queries given in the following format.

- You are given integers L, R, and X. Find the number of elements among $A_L, \ldots, A_R$ whose values are equal to X.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq A_i \leq N$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq L \leq R \leq N, 1 \leq X \leq N$, for each query.
- All values in input are integers.

How to solve this problem if a[i] <= 10^9?

https://atcoder.jp/contests/abc186/tasks/abc186_d

Given are N integers $A_1, \ldots, A_N$.

Find the sum of $|A_i - A_j|$ over all pairs i, j such that $1 \leq i < j \leq N$.

In other words, find $\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} |A_i - A_j|$.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $|A_i| \leq 10^8$
- A_i is an integer.

How to solve this problem if we had two arrays, and required |A[i] - B[j]| ?